

MLB ABS Challenge Run Value

Frozen 2026 regular-season snapshot through August 25

Research pipeline output

2026-08-26

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Provenance and validation

The analysis joins MLB live feeds, daily Statcast pitch exports, and Savant’s ABS detail. Raw responses are cached with SHA-256 manifests. A challenge is published only when its identity and explicit outcome reconcile to both the game’s final successful/failed totals and Savant.

Statcast alone is not the challenge ledger. Challenge timing and explicit overturned/upheld outcomes come from MLB’s MJ review records; Statcast provides pitch geometry and the pre-pitch state. Initial and corrected post-call states are generated from that common state, and Savant independently validates the event identity, original call, outcome, and geometric ruling. The full acquisition and reconciliation history is recorded in [docs/data-acquisition-walkthrough.md](#).

Table 1: Publication acceptance checks

check	passed	observed	expected
challenge_resolution	TRUE	8538	8538
feed_total_reconciliation	TRUE	8538	8538
final_inventory_reconciliation	TRUE	3634	3634
savant_challenge_match	TRUE	8537	8537
feed_only_challenge_quarantine	TRUE	1	1
geometry_agreement	TRUE	8537	8537
manual_stratified_audit	TRUE	50	50

scope	team_id	candidates	covered	coverage
league	NA	292381	292127	0.9991

scheduled_games	observed_games	coverage
7289	7289	1

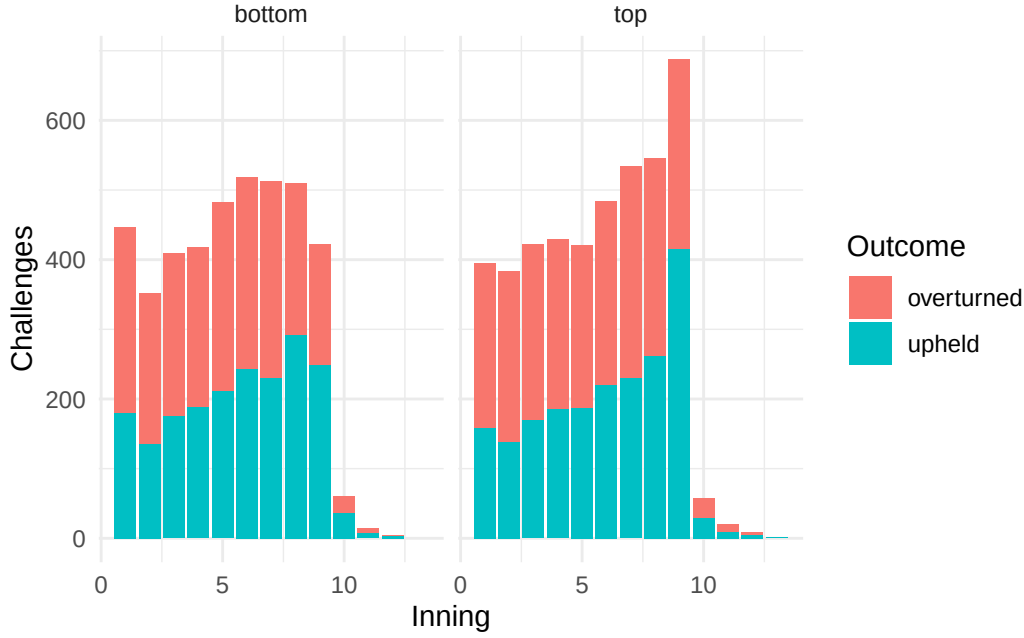
Table 4: Model validation diagnostics

diagnostic	n	metric_1	metric_2
RE288: 2025 holdout	712528	RMSE 0.9614	MAE 0.6235
Home WP: 2025 holdout	712528	Brier 0.1577	log loss 0.4700
RE288 vs Savant overturns	4577	RMSE 0.0496	correlation 0.9662

Challenge timing and success-rate accounting

Explicit **false** outcomes are upheld challenges, not missing observations. The denominator below is therefore every resolved attempt.

attempts	overturned	upheld	unresolved	success_rate
8537	4577	3960	0	0.5361



Actual challenge value

Immediate gain is the count-aware corrected-state minus original-state value from the challenger's perspective. Upheld calls have zero immediate gain and their modeled marginal inventory cost is reported separately.

Table 6: League challenge value

at-tempts	over-turned	up-held	suc-cess_pct	gross_re	failed_inven-tory_re	net_re	gross_wpa	failed_inven-tory_wpa
8537	4577	3960	53.614	833.56	14.653	818.907	17.893	0.171

Table 7: Per-failure benchmark versus modeled inventory cost

upheld	Savant fixed RE	modeled inventory RE	modeled inventory WPA
3960	0.21	0.0037	0

Missed and out-of-challenges value

These are realized correctable calls. The out-of-challenges column does not assume that a club would have challenged every such pitch.

Table 8: Ten teams with the most realized RE lost at zero inventory

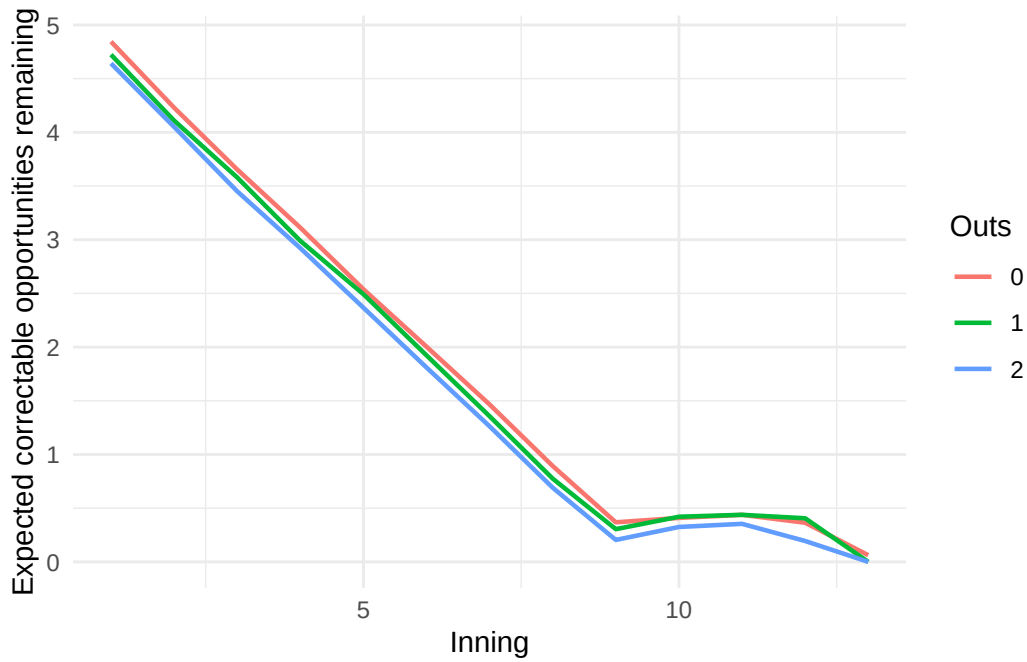
Team	Missed available	Missed RE	Out of challenges	Out RE
WSH	495	60.857	69	12.762
CWS	405	52.224	85	12.279
MIN	392	48.390	74	12.042
HOU	410	50.000	62	10.817
CLE	438	52.200	72	10.256
MIL	526	64.708	69	10.236
SF	416	47.229	62	8.716
NYM	436	56.930	55	8.566
TB	454	57.408	49	8.113
PIT	494	64.429	57	7.570

Remaining-opportunity inventory curves

Inventory is expressed first in future correctable opportunities, then RE and WPA. Curves use leave-one-game-out state averages; marginal values use the observed-behavior recursion.

Table 9: League-average remaining opportunity at game start

Correctable	Offense	Defense	Expected RE	Expected WPA	Unit 1 RE	Unit 2 RE
4.9987	2.6726	2.3261	0.6993	0.0101	0.0106	0.0067



Team comparisons and WPA context

Raw totals and opportunity-adjusted attempt residuals are shown together; neither is interpreted causally.

Table 10: Challenge outcomes and net value by team

team	attempts	overturns	upheld	success_pct	net_re	net_wpa
CIN	309	193	116	62.460	37.699	0.756
ATH	304	177	127	58.224	37.630	0.639
MIN	363	197	166	54.270	35.792	0.828
COL	320	170	150	53.125	33.415	0.745
CWS	353	170	183	48.159	32.454	1.061
LAA	302	162	140	53.642	31.715	0.424
MIA	324	183	141	56.481	29.954	0.945
NYY	331	181	150	54.683	29.884	0.540
NYM	319	165	154	51.724	29.457	0.849
CHC	266	156	110	58.647	29.094	0.385
BOS	261	142	119	54.406	29.067	0.631
SEA	265	135	130	50.943	26.499	0.935
PHI	272	155	117	56.985	26.302	0.362
HOU	312	167	145	53.526	25.970	0.462

team	attempts	overturns	upheld	success_pct	net_re	net_wpa
TEX	240	137	103	57.083	25.967	0.531
DET	262	154	108	58.779	25.879	0.789
MIL	306	153	153	50.000	25.680	0.449
AZ	244	148	96	60.656	25.676	0.401
KC	277	153	124	55.235	25.641	0.654
SD	269	140	129	52.045	25.045	0.810
LAD	255	142	113	55.686	24.980	0.477
BAL	262	132	130	50.382	24.851	0.464
TOR	276	148	128	53.623	24.303	0.529
STL	229	118	111	51.528	23.774	1.125
ATL	247	130	117	52.632	23.596	0.437
SF	265	131	134	49.434	22.371	0.306
PIT	286	132	154	46.154	22.036	0.399
CLE	275	130	145	47.273	21.880	0.239
WSH	275	135	140	49.091	21.595	0.436
TB	268	141	127	52.612	20.703	0.111

Table 11: Missed, zero-inventory, and remaining-opportunity value

Team	Missed RE	Zero-inventory RE	Expected left	Unit 1 RE	Unit 2 RE
WSH	60.857	12.762	4.995	0.011	0.007
CWS	52.224	12.279	4.995	0.011	0.007
MIN	48.390	12.042	4.998	0.011	0.007
HOU	50.000	10.817	4.988	0.010	0.006
CLE	52.200	10.256	5.004	0.011	0.006
MIL	64.708	10.236	4.995	0.012	0.007
SF	47.229	8.716	5.007	0.010	0.006
NYM	56.930	8.566	5.006	0.011	0.007
TB	57.408	8.113	4.992	0.011	0.006
PIT	64.429	7.570	5.001	0.010	0.007
ATL	48.631	7.542	5.003	0.010	0.006
BAL	60.855	7.522	4.990	0.010	0.007
LAA	63.438	7.358	4.994	0.012	0.008
SEA	58.343	7.300	4.992	0.012	0.007
NYN	42.961	7.235	5.007	0.008	0.006
KC	56.943	7.110	4.997	0.011	0.007
COL	50.048	7.041	5.011	0.011	0.007
MIA	54.296	6.895	4.996	0.010	0.007
ATH	61.298	6.204	5.004	0.011	0.008
SD	58.435	5.510	5.004	0.011	0.006

Team	Missed RE	Zero-inventory RE	Expected left	Unit 1 RE	Unit 2 RE
CHC	58.888	5.311	5.006	0.010	0.007
TOR	45.314	4.986	4.996	0.008	0.005
BOS	62.779	4.897	4.999	0.013	0.007
DET	53.881	4.582	5.000	0.010	0.006
CIN	65.826	4.485	4.998	0.012	0.009
PHI	55.690	3.772	4.995	0.010	0.006
TEX	60.956	3.771	4.996	0.012	0.007
LAD	61.222	3.370	4.995	0.011	0.007
STL	61.526	3.007	5.004	0.010	0.006
AZ	57.843	2.798	4.994	0.009	0.006

Table 12: Opportunity-adjusted attempts with 95% game-cluster intervals

team	observed_at- tempts	expected_at- tempts	challenge_residual	lower	upper
MIN	363	268.554	94.446	59.654	135.439
CWS	353	264.591	88.409	57.239	127.888
COL	320	243.145	76.855	39.253	115.998
NYN	331	275.217	55.783	21.957	91.240
MIA	324	279.465	44.535	15.823	72.202
NYM	319	279.310	39.690	7.670	71.745
HOU	312	282.551	29.449	-2.140	61.176
ATH	305	288.001	16.999	-11.163	44.938
LAA	302	289.692	12.308	-17.617	42.532
MIL	306	297.153	8.847	-22.685	39.798
PIT	286	280.138	5.862	-24.029	39.697
TOR	276	271.576	4.424	-28.520	34.470
TB	268	267.339	0.661	-27.499	32.743
SD	269	276.530	-7.530	-35.433	25.654
BOS	261	271.229	-10.229	-42.964	18.819
SEA	265	276.154	-11.154	-43.290	19.736
CLE	275	287.460	-12.460	-45.088	19.710
SF	265	279.730	-14.730	-45.412	15.174
WSH	275	294.865	-19.865	-49.811	11.184
PHI	272	293.278	-21.278	-49.850	7.738
ATL	247	268.298	-21.298	-54.886	12.246
LAD	255	276.751	-21.751	-50.664	7.188
CIN	309	330.860	-21.860	-54.339	13.766
DET	262	290.756	-28.756	-56.707	-0.243
BAL	262	291.601	-29.601	-59.644	2.466

team	observed_at- tempts	expected_at- tempts	challenge_residual	lower	upper
KC	277	306.624	-29.624	-60.095	-1.748
TEX	240	284.727	-44.727	-76.582	-14.241
CHC	266	313.755	-47.755	-78.628	-15.894
AZ	244	293.980	-49.980	-82.836	-20.980
STL	229	317.449	-88.449	-123.421	-56.728

Table 13: Top 15 teams by net challenge RE with 95% game-cluster intervals

Team	Net RE	Lower	Upper
CIN	37.699	27.918	49.089
ATH	37.630	29.725	46.439
MIN	35.792	26.266	45.504
COL	33.415	24.779	44.277
CWS	32.454	22.982	41.866
LAA	31.715	23.570	41.096
MIA	29.954	23.131	37.606
NYN	29.884	22.398	37.959
NYM	29.457	22.329	38.908
CHC	29.094	20.723	37.766
BOS	29.067	21.732	37.745
SEA	26.499	18.443	35.570
PHI	26.302	19.100	33.949
HOU	25.970	19.282	33.141
TEX	25.967	19.623	32.978

Sensitivity and limitations

The boundary sensitivity removes pitches within 0.25 inches of the official ABS edge. This targets public-tracking and rounding uncertainty without reclassifying the remaining pitches.

correctable_all	correctable_outside_boundary	out_all	out_outside_boundary
19757	16005	1428	1156

Unchallenged ABS outcomes remain geometry-based inferences. The RE288 and home-win GAM are empirical 2023–2025 models, and all adjusted comparisons may retain unmeasured differences in communication, pitcher/catcher confidence, and game context.